

1) Suppose a large local bakery sells cupcakes, and historically, 20% of the cupcakes sold are chocolate-flavored. Imagine taking an SRS, from all the cupcakes they make in a day, of 50 cupcakes from the bakery.

(a) What is the mean of the sampling distribution of $\hat{p}$? Why?

(b) Find the standard deviation of the sampling distribution of $\hat{p}$. Check the 10% condition, and explain how it is a reasonable assumption.

(c) Is the sampling distribution of $\hat{p}$ approximately Normal?

(d) If the sample size were 200 rather than 50, how would this change the sampling distribution of $\hat{p}$?

The center and shape would not be affected, but the standard deviation would change from:

2) A fast-food chain claims that 85% of its drive-thru orders are completed within 3 minutes. To verify this claim, a manager selects a random sample of 150 drive-thru orders from the past week for analysis. Among these, 123 orders were completed within 3 minutes.

(a) If the company's claim of completing 85% of orders within 3 minutes is accurate, what is the probability that the proportion in an SRS of 150 orders is 0.82 or less? *Check your inference conditions and show all work!*

(b) A critic argues, "Your sample shows 82% of orders completed within 3 minutes, which is lower than the claimed 85%. Therefore, your claim of 85% completion rate is wrong!" Explain why the probability calculation in (a) does not necessarily refute the company's claim of an 85% completion rate.

3) The National Health Survey (NHS) reports that 80% of adults in a certain region engage in regular physical exercise. To determine if this trend holds true in a specific neighborhood, a local health organization conducts a survey of a simple random sample of 150 adults, in a neighborhood with over 2000 adults. Among these adults, 115 report engaging in regular physical exercise.

(a) What is the probability that we get this result or lower, if we assume that the NHS's result holds true in our neighborhood?

Communicate:

Conditions:

Calculate:

Conclude:

(b) Does your conclusion suggest that the proportion of adults in the specific neighborhood who engage in regular physical exercise differs significantly from the 80% reported by the National Health Survey? In your opinion, what probability would you have to get in order for the results to seem significant?